

Simulating Urban Atlas

Can OSM be used as a source for landuse/landcover?

Jerry Clough (SK53)

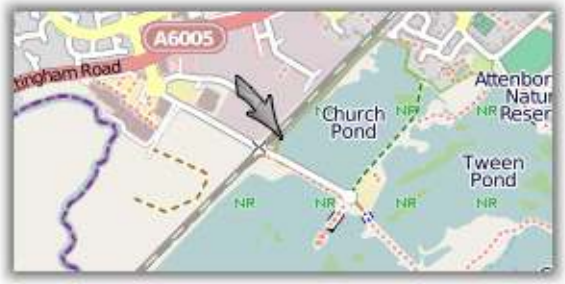


Landuse mapping in OSM

- Mainly import driven
  - Corine
  - US States (GA, NJ)
- Imports as a base for modification
  - But are they?
- Enhance cartographic rendered outputs
- Are they useful?

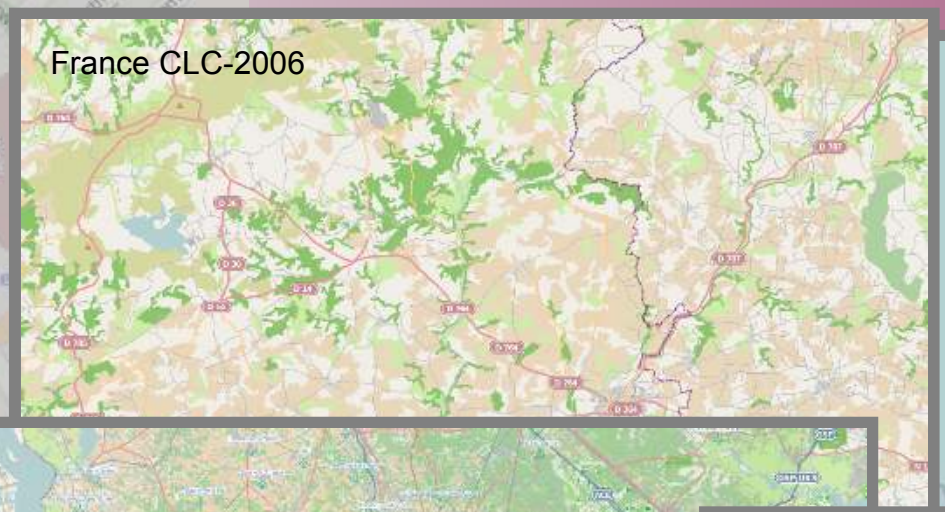
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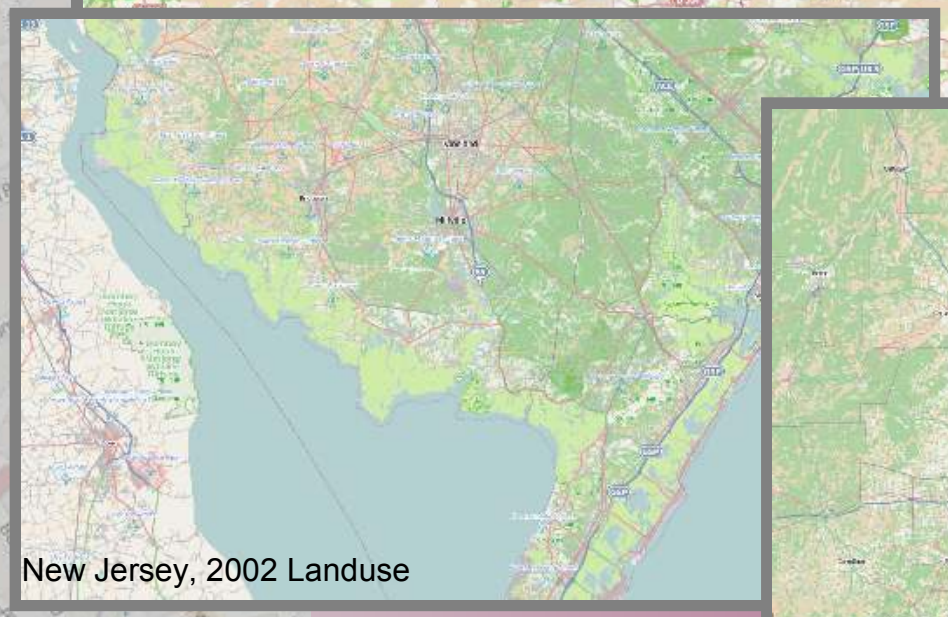


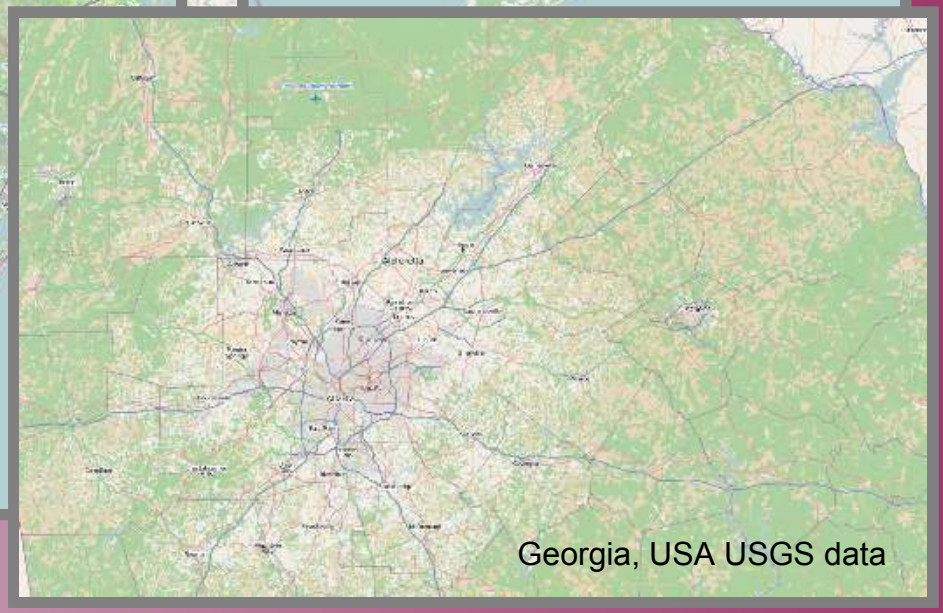
Your first Node in OSM, do you remember?
(Thu Dec 11 10:23:20 CET 2008)

OSM Landuse Imports













CLC lacks detail & precision : Spain



CLC lacks detail & precision : France



Use-cases for land-use

- Environmental
  - Hydrology
  - Pollution
  - Ecological
  - Sustainable resources
- Planning
  - NIMBY toolkit

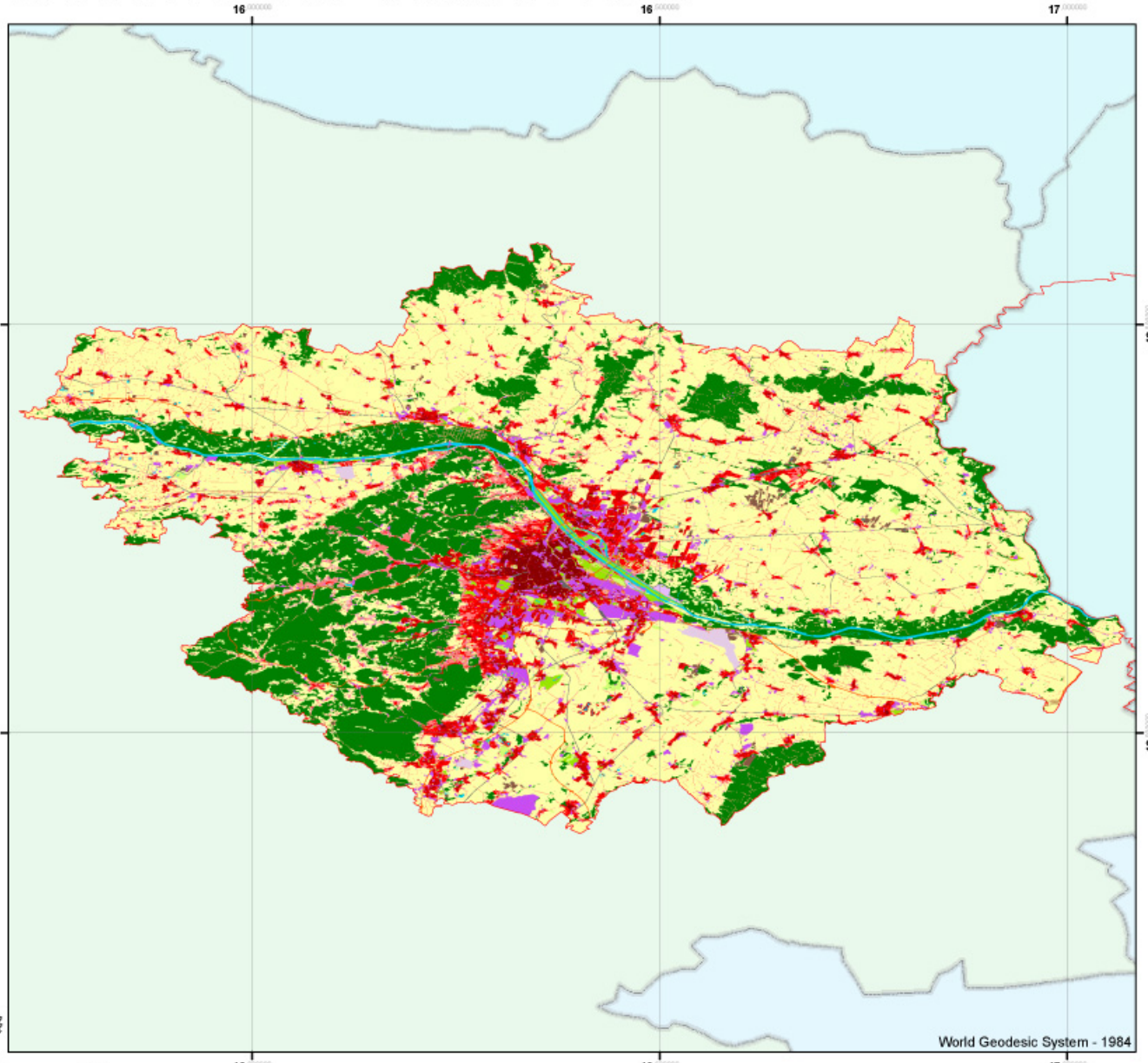


Urban Atlas

- 300+ EU cities population >100k
  - 119 in April 2010
  - 228 in Sept. 2010
- Baseline date 2006-7
- Used 2.5 m imagery
- 5-6 year refresh cycle
- Minimum Map Unit (MMU) 0.25 ha urban / 1 ha rural

http://sia.eionet.europa.eu/Land Monitoring Core Service/Urban Atlas

URBAN ATLAS - AT001L-Wien





Legend

- 11100: Continuous Urban fabric (S.L. > 80%)
- 11210: Discontinuous Dense Urban Fabric (S.L.: 50% - 80%)
- 11220: Discontinuous Medium Density Urban Fabric (S.L.: 30% - 50%)
- 11230: Discontinuous Low Density Urban Fabric (S.L.: 10% - 30%)
- 11240: Discontinuous very low density urban fabric (S.L. < 10%)
- 11300: Isolated Structures
- 12100: Industrial, commercial, public, military and private units
- 12210: Fast transit roads and associated land
- 12220: Other roads and associated land
- 12230: Railways and associated land
- 12300: Port areas
- 12400: Airports
- 13100: Mineral extraction and dump sites
- 13300: Construction sites
- 13400: Land without current use
- 14100: Green urban areas
- 14200: Sports and leisure facilities
- 20000: Agricultural, semi-natural and wetland areas
- 30000: Forest
- 50000: Water

déc 03, 2009

SIRS



Opportunity

- Urban Atlas
  - Scale (~1:10k) ++ cf. with OSM
  - Discrete areas
  - Urban focus
  - Detail (small MMU size)
- Good chance to examine land-use mapping in OSM
  - Objective comparison to external data
  - Produce equivalent outputs
  - Learn more about :
    - Accuracy/Applicability/Currency/Consistency

UA to OSM Category Mapping 1

UA Code	UA Description	OSM Tags	Comments
11100 11110 11120 11130 11140	Urban Fabric Continuous /Discontinuous Urban Fabric	`landuse=residential`	There are no widely used sub-classes, certainly none which correspond with the density grouping of UA. See detailed discussion below.
11300	Isolated Dwellings	`landuse=farmyard`	Other isolated houses would need to be identified computationally.

12100	Industrial and Commercial land	`landuse=retail`
		`landuse=commercial`
		`landuse=industrial`
		`amenity=university`
		`amenity=hospital,amenity=school`
			For campus sites (education and health) it is assumed that green spaces (parks, sports pitches, woodland, water, etc) are handled by their respective tags.

12210	Fast transit roads	`highway=motorway, motorway_link`	Motorways buffered 30 m

12220	Other roads	`highway=trunk, trunk_link, primary, primary_link`
		`highway=secondary, secondary_link`
		`highway=tertiary, tertiary_link`
		`highway=unclassified, residential, pedestrian`
			Primary and Trunk buffered 20 m
			Secondary roads buffered to 10 m
			Tertiary roads buffered to 10m
			other roads buffered to 7.5m

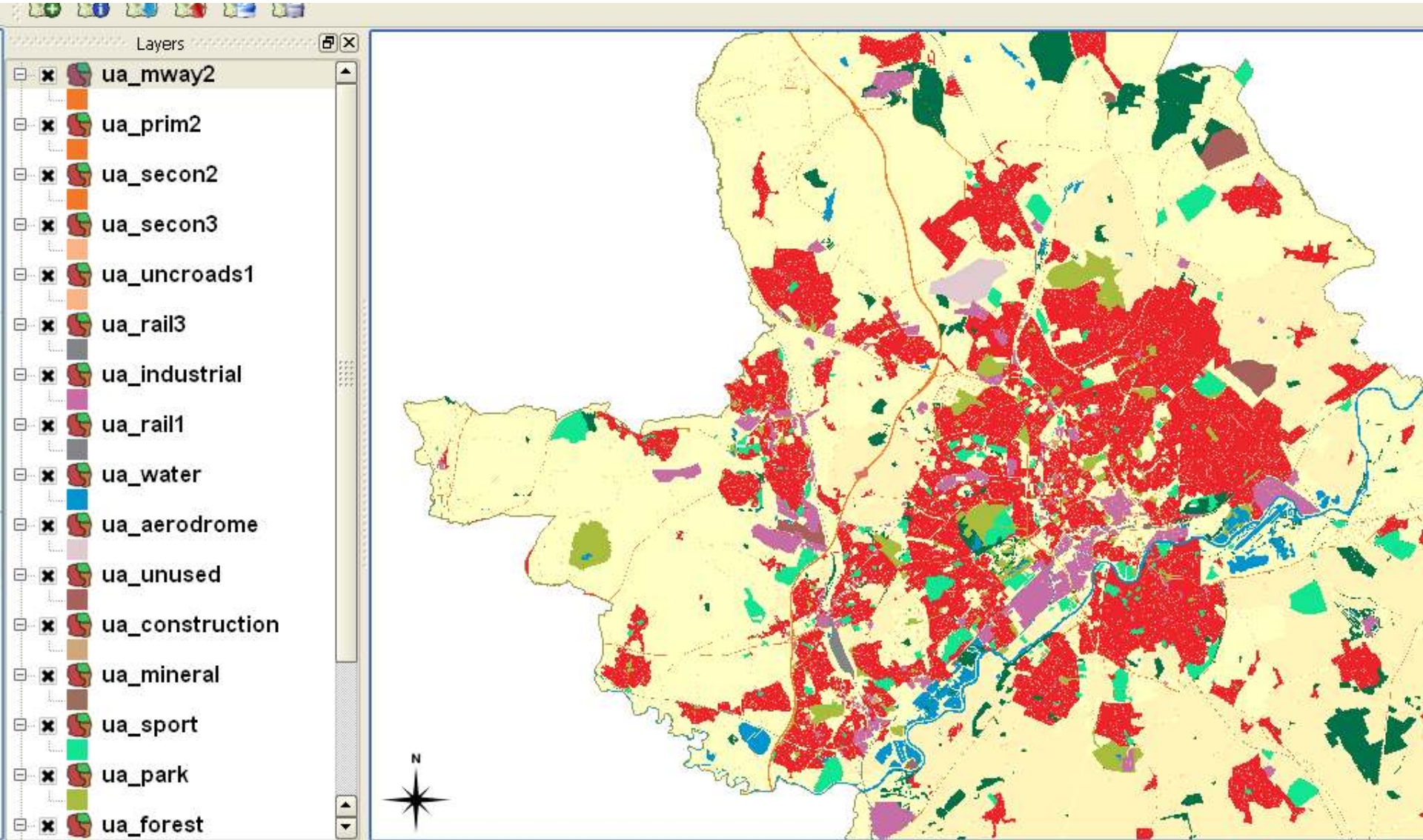
UA to OSM Category Mapping 2

UA Code	UA Description	OSM Tags	Comments
12230	Railways	`landuse=railway` `railway=rail, preserved`	Trams were not included even though one runs in a railway corridor. Rail buffered to 10m
12300	Port		Not included in this study.
12400	Airfields	`aeroway=aerodrome`	
13100	Quarries and Landfill	`landuse=quarry` `landuse=landfill`	
13300	Construction	`landuse=construction`	
13400	Unused Land	`landuse=greenfield` `landuse=brownfield`	

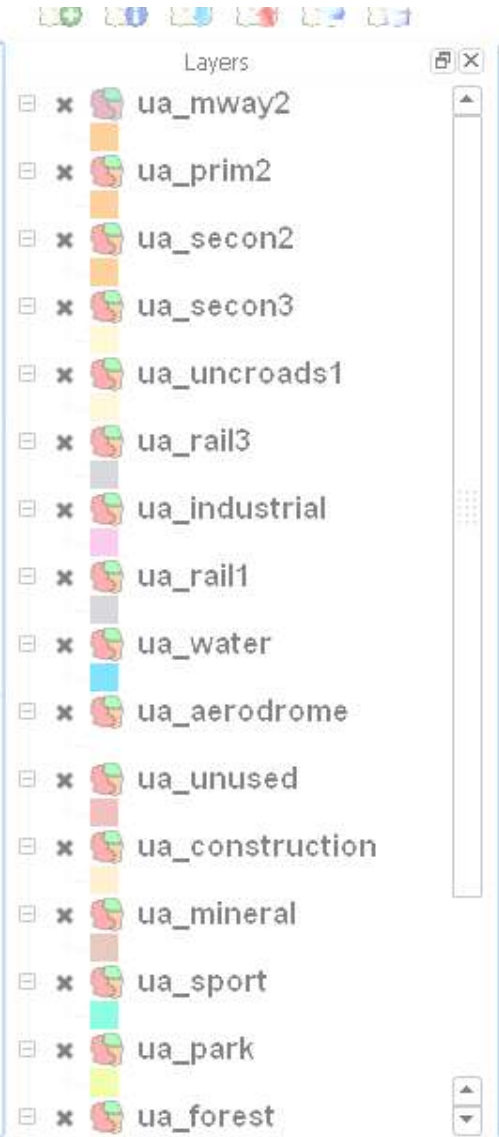
UA to OSM Category Mapping 3

UA Code	UA Description	OSM Tags	Comments
14100	Parks, Urban Green Space	amenity=graveyard landuse=cemetery leisure=park leisure=village_green	
14200	Sports Areas	landuse=allotments landuse=recreation_ground leisure=golf_course leisure=pitch leisure=stadium	
20000	Agricultural Land	landuse=farm landuse=farmland landuse=pasture landuse=orchard landuse=vineyard leisure=nature_reserve natural=scrub,natural=heath natural=wetland natural=rock,natural=scree	Additional OSM tags are also valid for this code (e.g., natural=glacier)
30000	Woods & Forest	natural=wood landuse=forest	
50000	Water	landuse=reservoir waterway=riverbank natural=water	

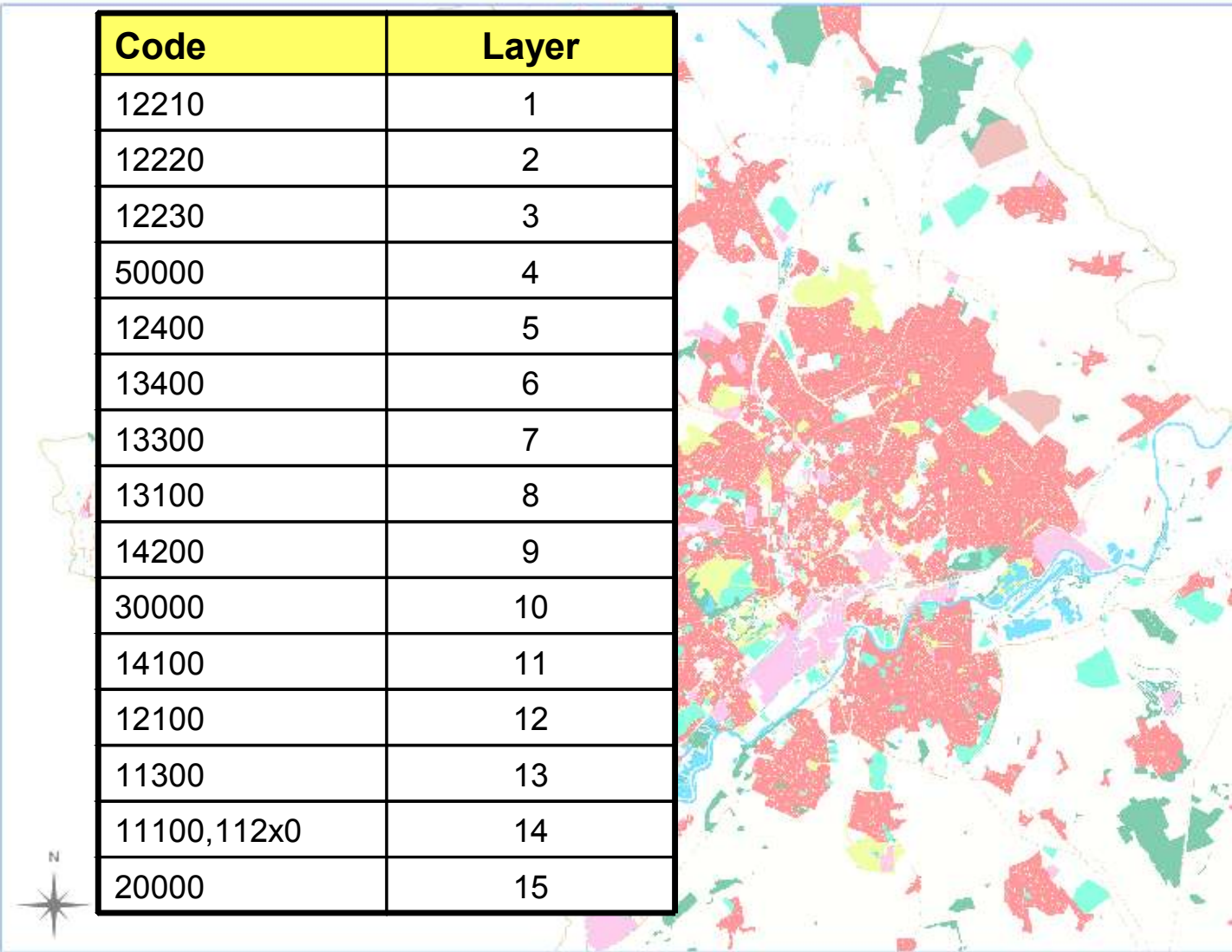
Painter's Algorithm in QGIS



Painter's Algorithm in QGIS

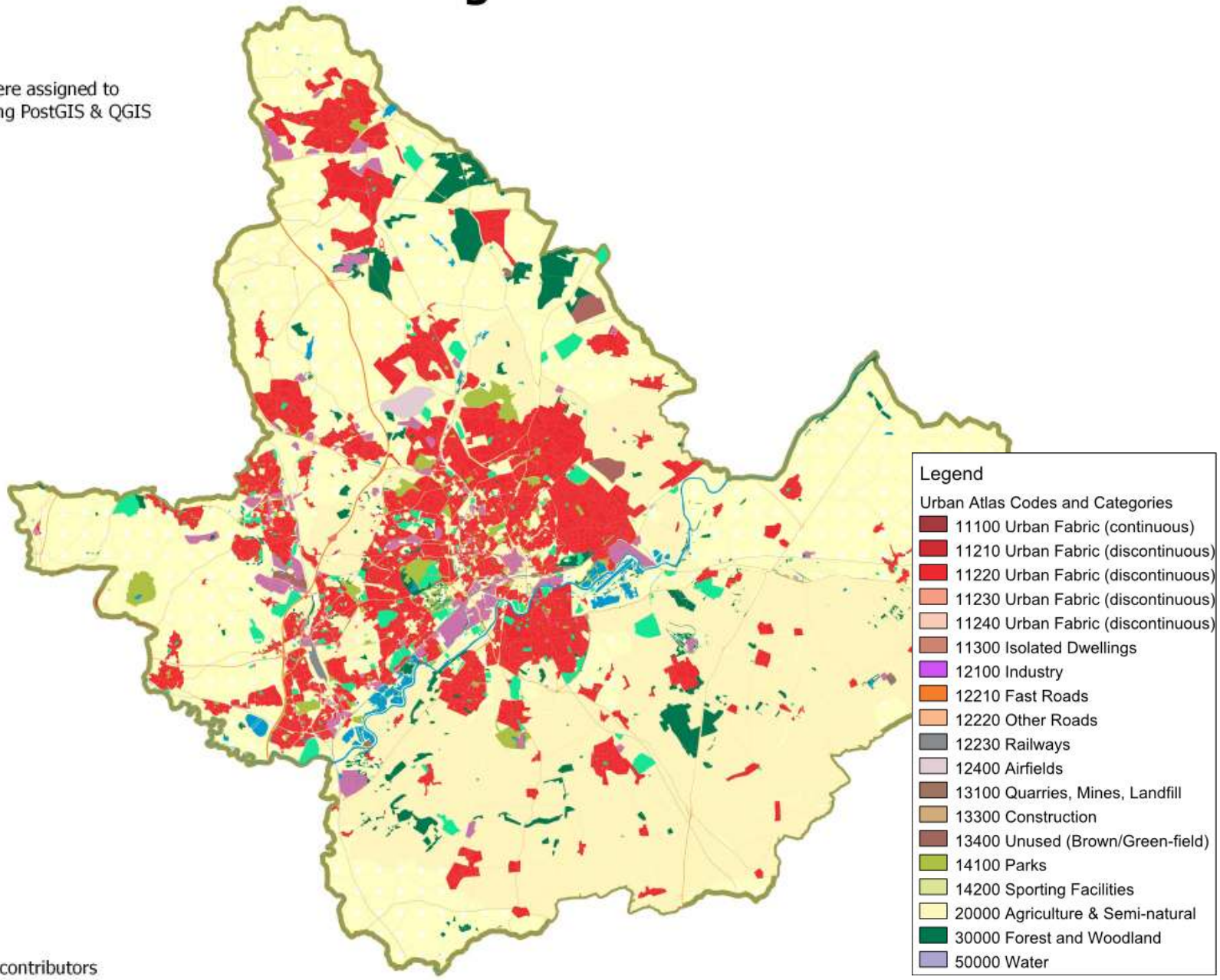


Code	Layer
12210	1
12220	2
12230	3
50000	4
12400	5
13400	6
13300	7
13100	8
14200	9
30000	10
14100	11
12100	12
11300	13
11100,112x0	14
20000	15



Urban Atlas - UK029L -Nottingham

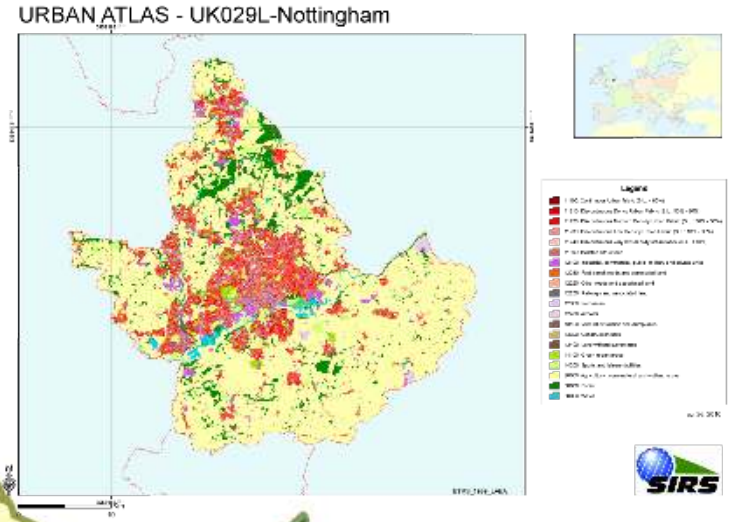
OpenStreetMap data were assigned to
Urban Atlas classes using PostGIS & QGIS



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Urban Atlas - UK029L -Nottingham

OpenStreetMap data were assigned to
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Legend

Urban Atlas Codes and Categories

- 11100 Urban Fabric (continuous)
- 11210 Urban Fabric (discontinuous)
- 11220 Urban Fabric (discontinuous)
- 11230 Urban Fabric (discontinuous)
- 11240 Urban Fabric (discontinuous)
- 11300 Isolated Dwellings
- 12100 Industry
- 12210 Fast Roads
- 12220 Other Roads
- 12230 Railways
- 12400 Airfields
- 13100 Quarries, Mines, Landfill
- 13300 Construction
- 13400 Unused (Brown/Green-field)
- 14100 Parks
- 14200 Sporting Facilities
- 20000 Agriculture & Semi-natural
- 30000 Forest and Woodland
- 50000 Water

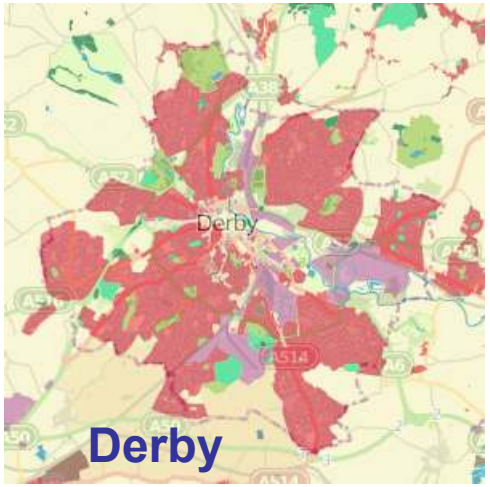
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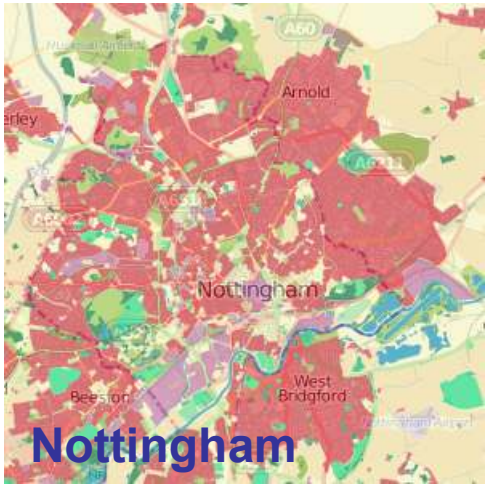
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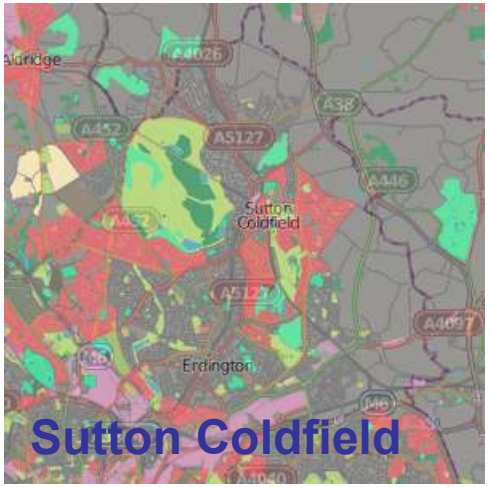
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- <Datasource>
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      WHEN highway IN ('trunk','trunk_link') THEN 10
      WHEN highway IN ('primary','primary_link') THEN 10
      WHEN highway IN ('secondary','secondary_link') THEN 7.5
      WHEN highway IN ('tertiary','tertiary_link') then 7.5
      WHEN
         railway IN ('rail','tram','preserved','narrow_gauge')
         THEN 10
      ELSE 3.75
      END),900913) as way
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      , railway
      , name
FROM planet_osm_line
WHERE (highway IN
      ('motorway','motorway_link' ,'trunk','trunk_link' ,'primary','
      primary_link' ,'secondary','secondary_link' ,'tertiary','tertia
      ry_link' ,'pedestrian','residential','unclassified'))
      OR (railway IN ('rail','tram','preserved','narrow_gauge')) )
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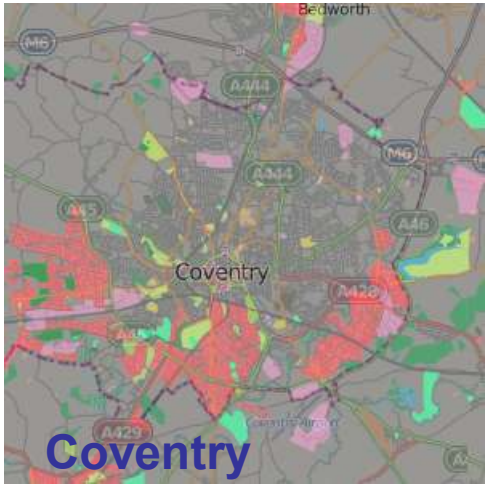
Mapnik Output

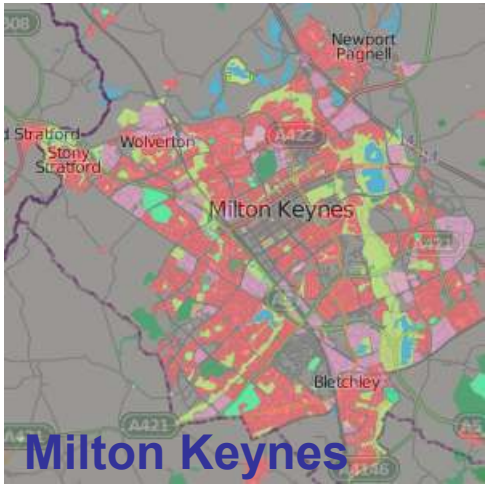




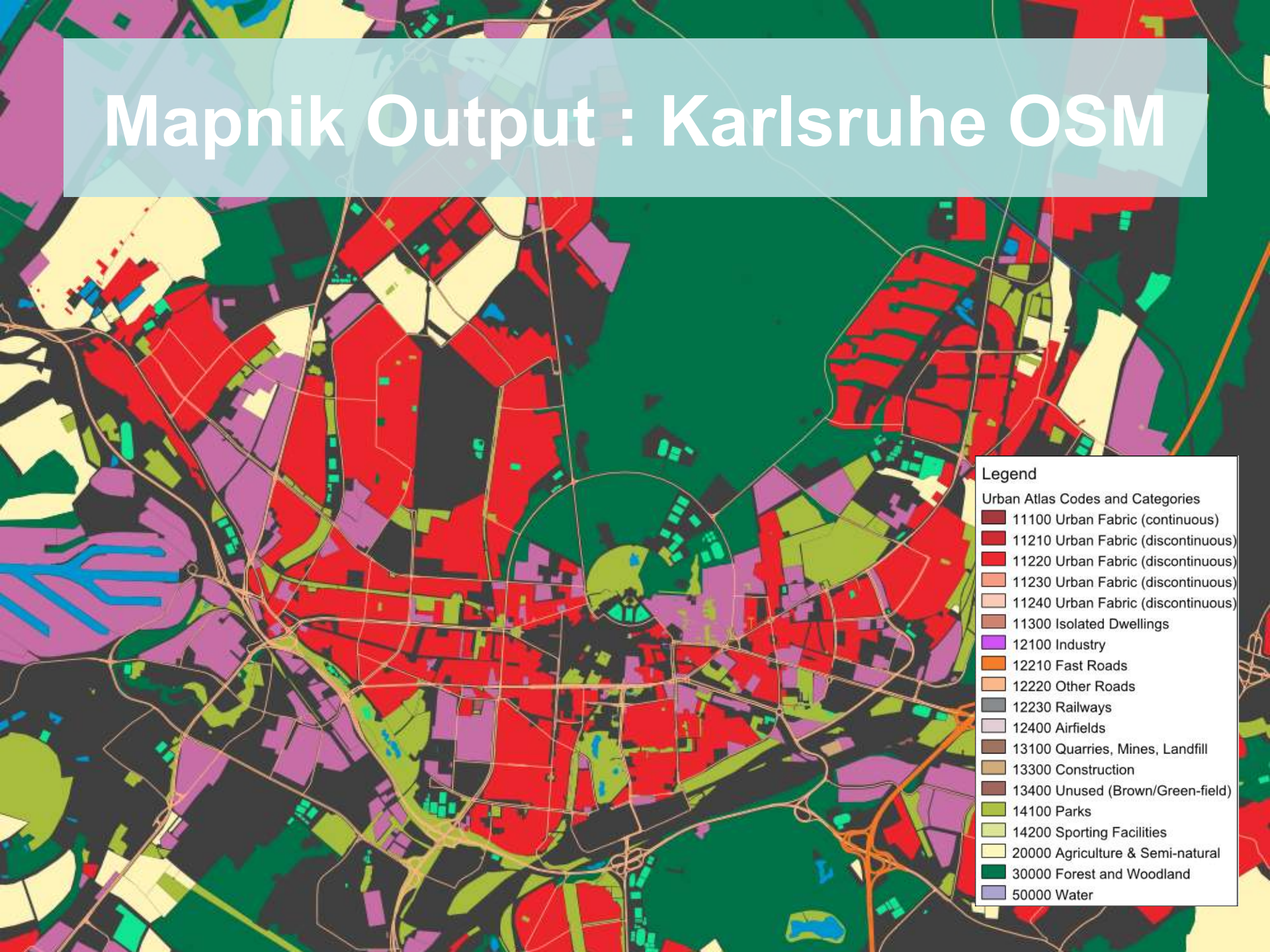








Mapnik Output : Karlsruhe OSM





BUT…

- Raster output only
  - No Shape file output
- Informational not Analytical
- Bad Polygons

→ **PostGIS**

The Problem with Polygons

- OSM
  - Broken polygons
  - Intersecting polygons
  - osm2pgsql
- PostGIS
  - Multipolygons
  - many set operations (UNION/Intersection)
- Essential tool: `cleangeometry` PostGIS function (SOGIS)

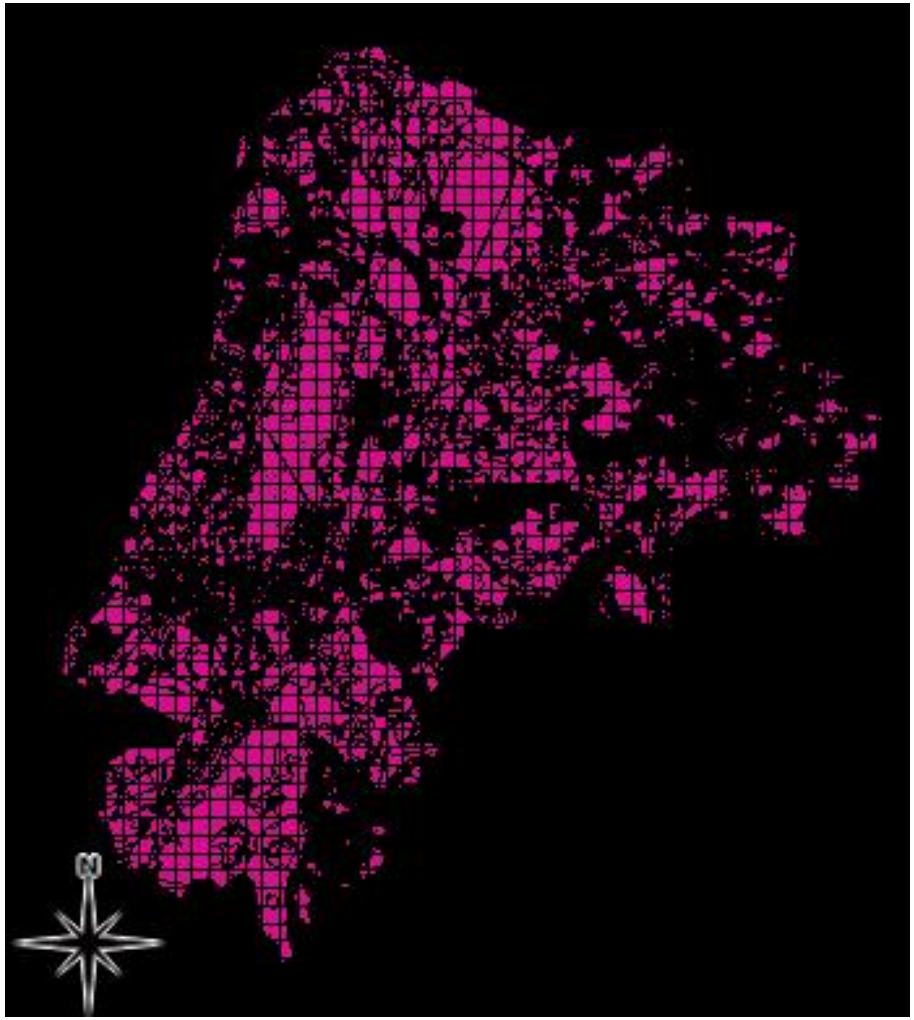
http://www.sogis1.so.ch/sogis/dl/postgis/cleanGeometry.sql



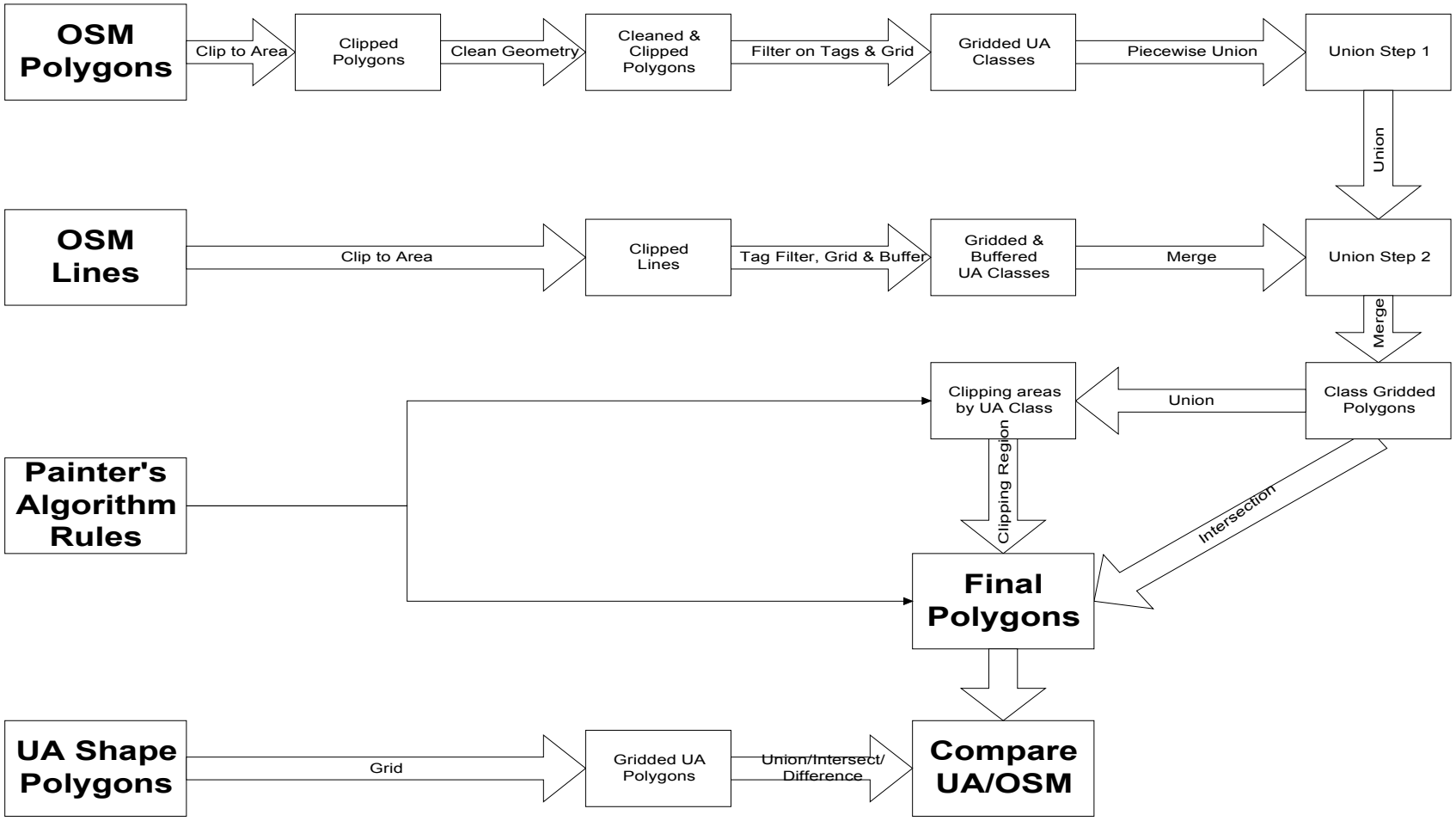


Gridded Output

- Intersection of all features on 1km grid
  - Reduce polygon size
  - Performance
  - Avoid joining on geometries (use key for grid cell)

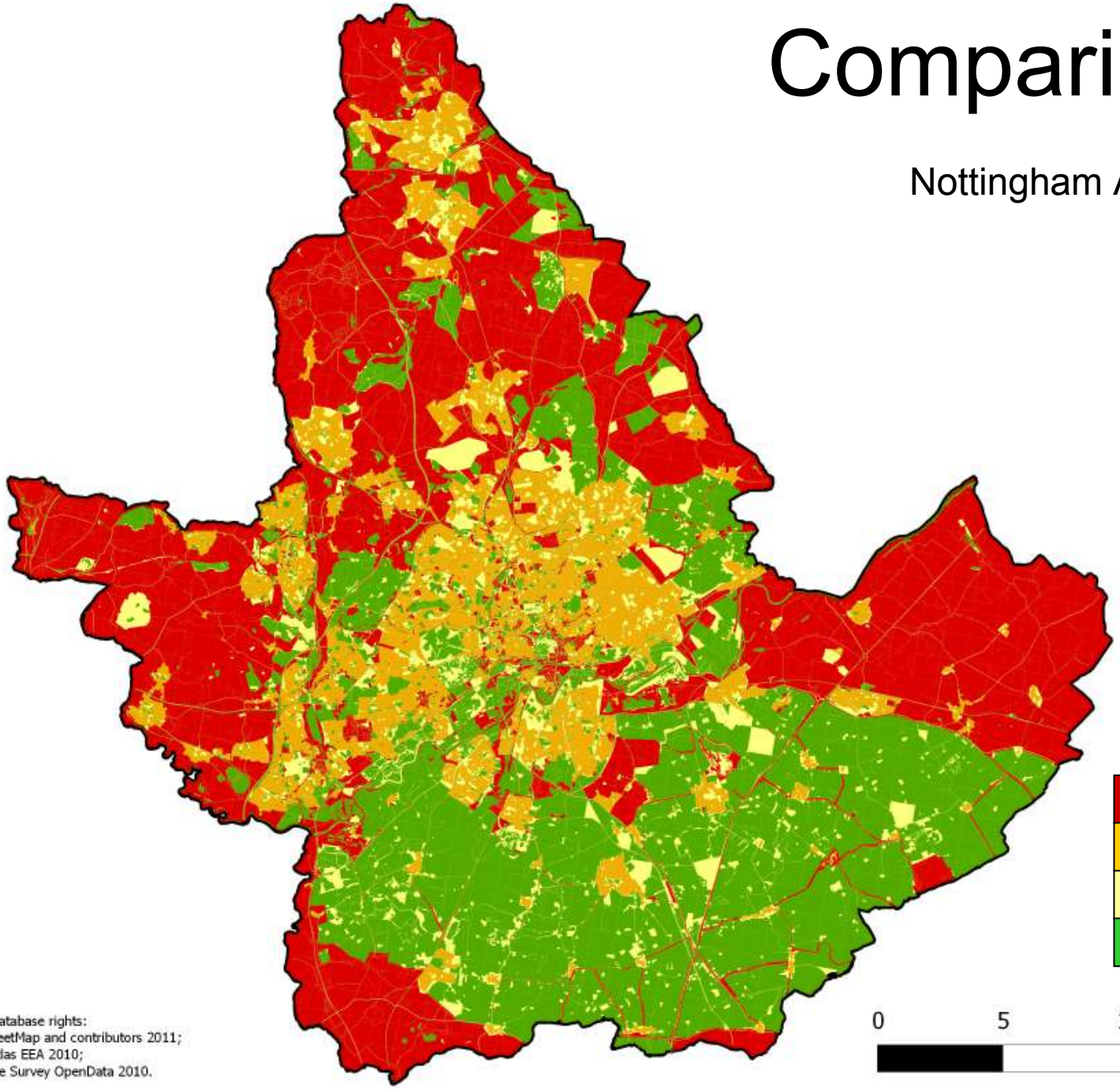


PostGIS Processing



Comparison 1

Nottingham Area

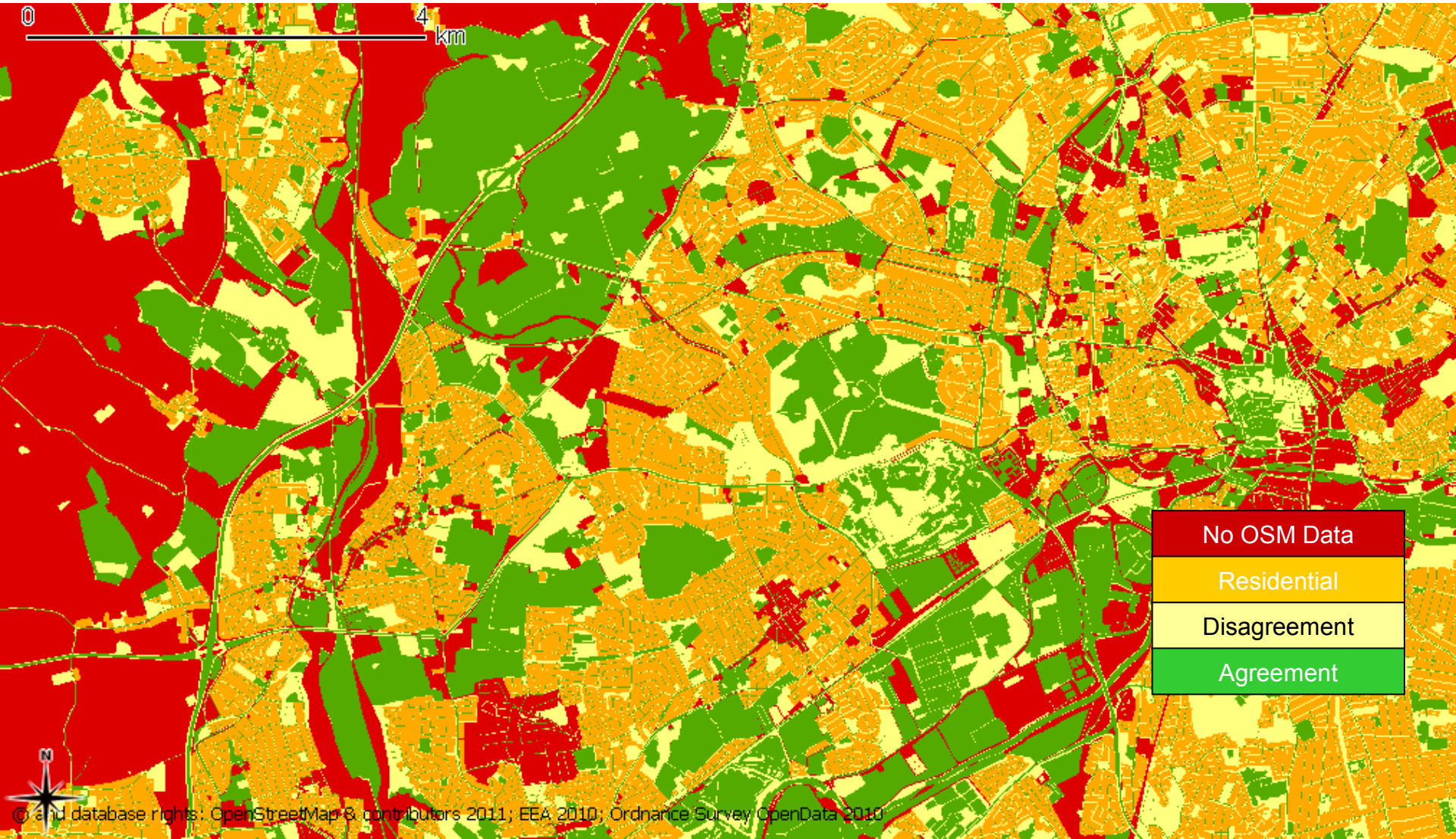


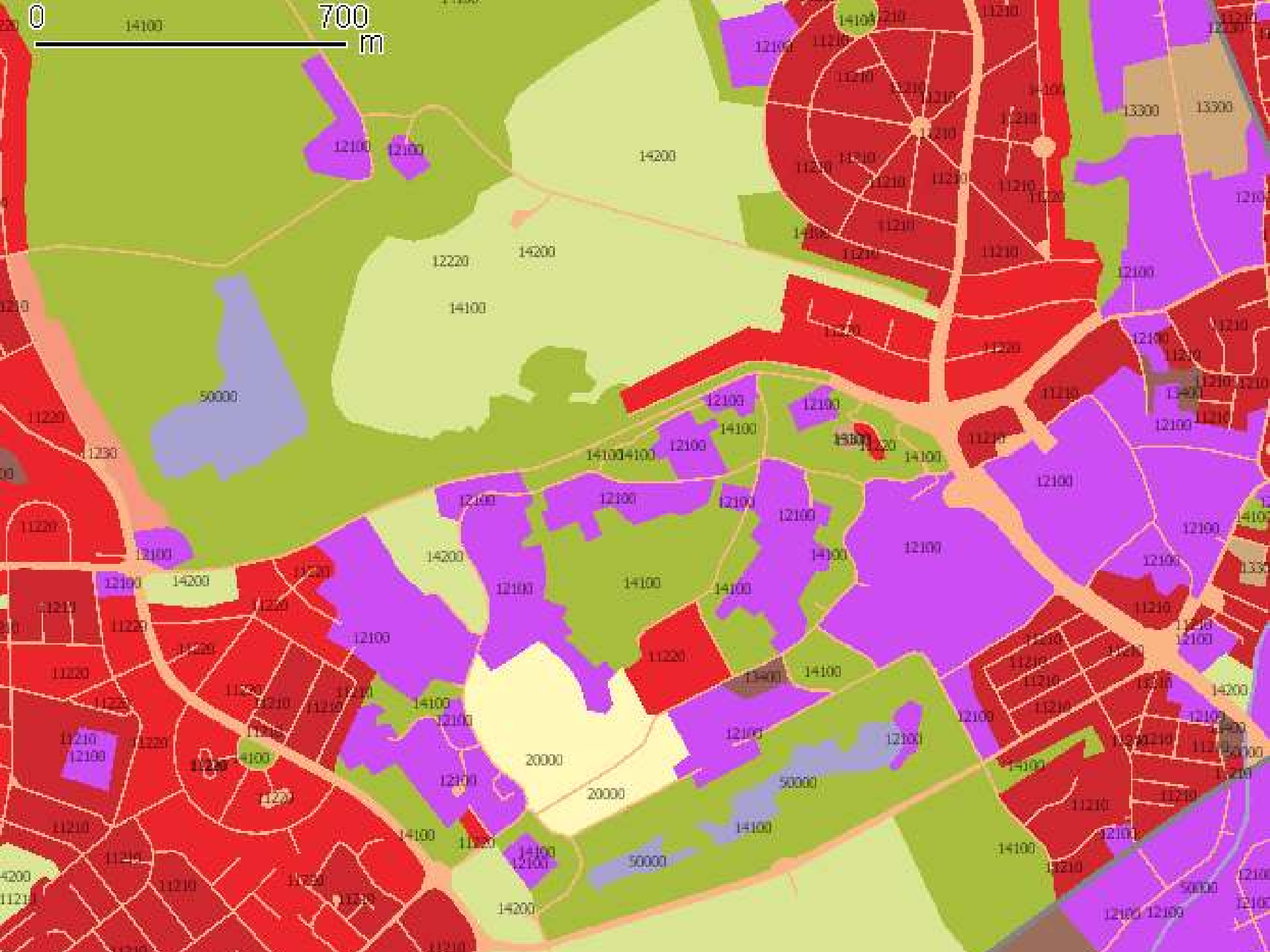
No OSM Data
Residential
Disagreement
Agreement

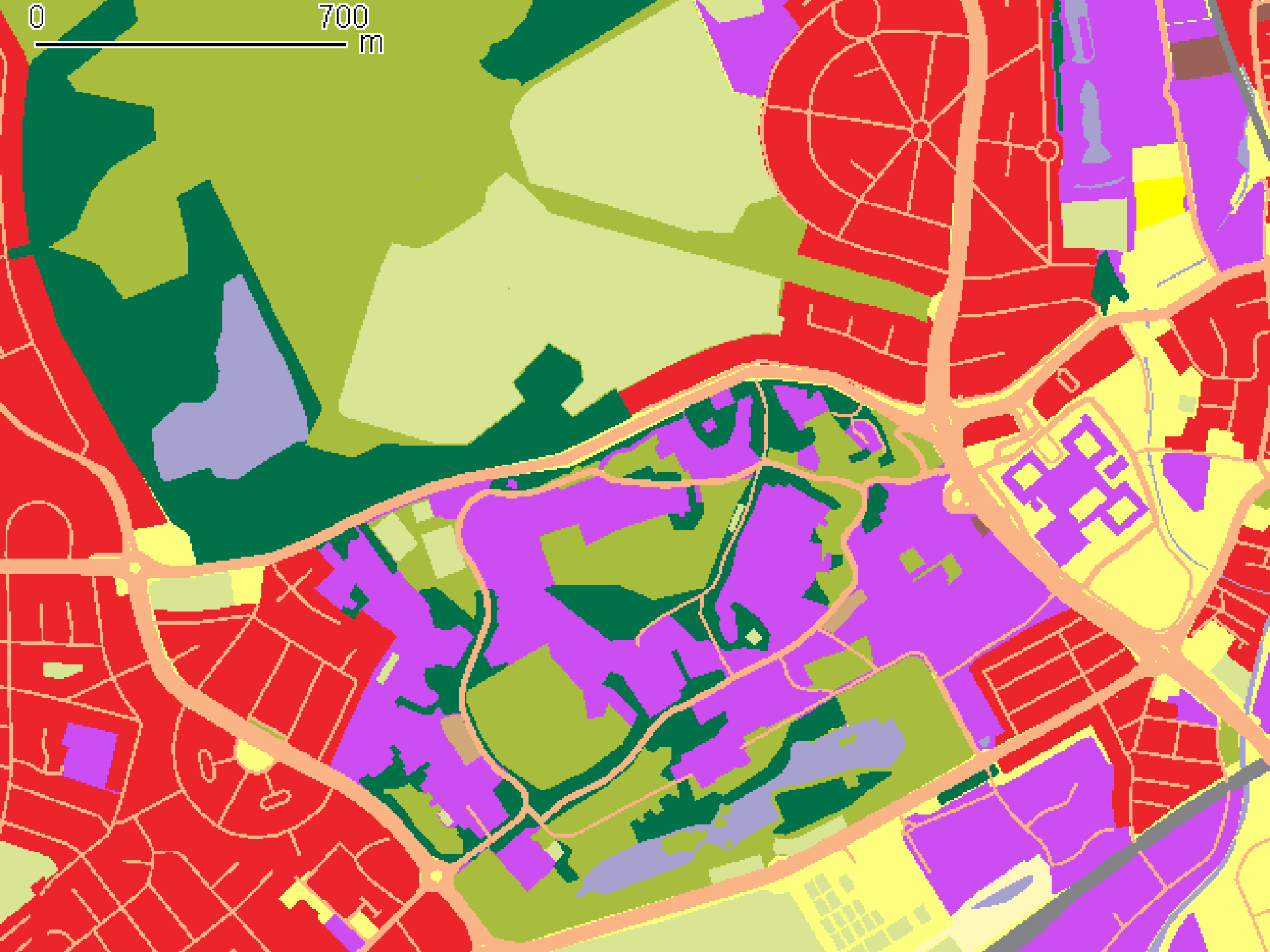
0 5 10 km

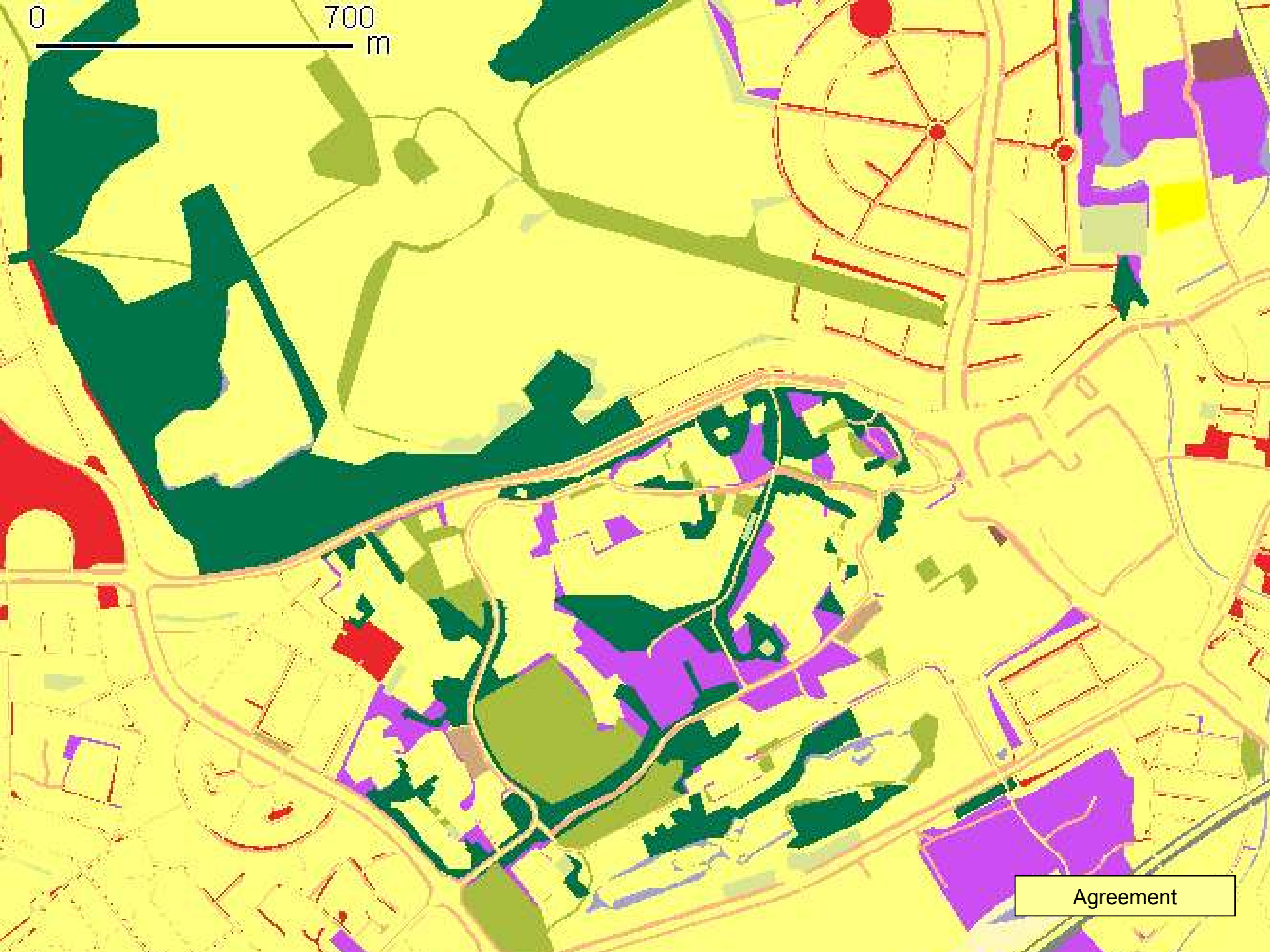
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Urban Atlas EEA 2010;
Ordnance Survey OpenData 2010.

Comparison 2




0
700
m


0
700
m


0
700
m
Agreement

Comparison: Numbers

	Area in hectares			% variance
UA Class	**UK029L (A)**	**Not in OSM (B)**	**OSM (C)**	**C %(A-B)**
11100,112x0	13430.9	1654.7	12822.2	109.00%
11300	271.6	167	55.6	53.00%
12100	5351.9	1856.8	2240.4	64.00%
12210	122.8	3.7	183.8	154.00%
12220	2923.8	420.5	3445.3	138.00%
12230	308.3	54.3	393.1	155.00%
12400	402.9	375.3	197.8	714.00%
13100	321	153.1	43.8	26.00%
13300	232.8	167	38.1	58.00%
13400	177.9	375.3	302.4	-153.00%
14100	1376.7	349.7	1187.9	116.00%
14200	3014.7	890.9	1752	82.00%
20000	56038.2	29784.8	25478.2	97.00%
30000	5490.6	2260.4	3208.7	99.00%
50000	904.6	111.3	903.9	114.00%


Arnold
Lambley
Nuthall
Bulwell
Gedling
Carlton
Bilborough
Strelley
NOTTINGHAM
Netherfield
Colwick
Wollaton
Wollaton Park
Holme Pierrepont
Bramcote
Wilford
West Bridgford
Gamston
Basingfield

Conclusions

- Crowd sourcing of land-use works
- Cartographic (raster) products are straightforward to produce
- Analytical (vector) products would benefit from more tool support
- Tagging can be enriched to provide finer granularity